1.List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Key Differences

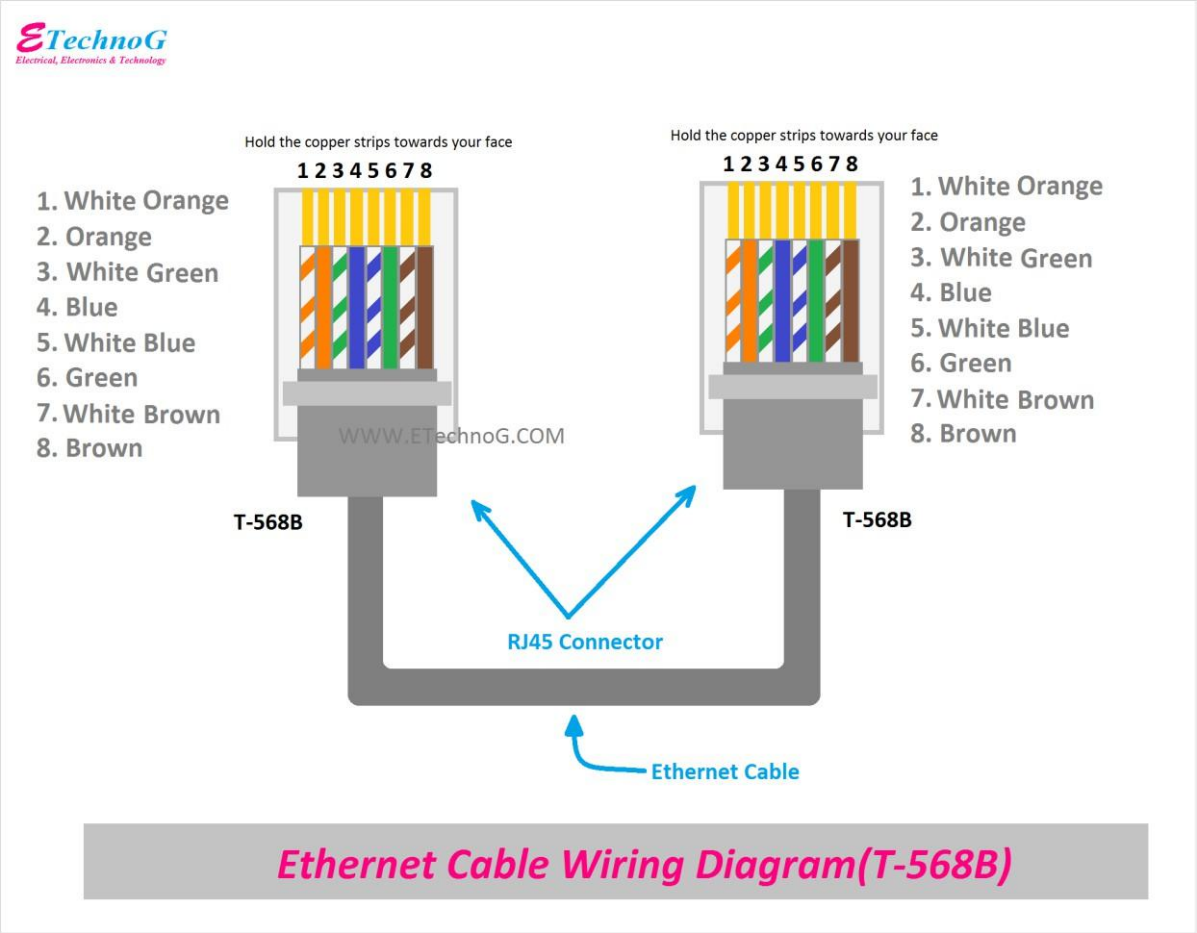
1. **Transmission Medium & Signal**
    - **Ethernet (Copper):** Uses copper wires to transmit data via electrical pulses.
    - **Fiber Optic:** Uses thin strands of glass or plastic to transmit data via pulses of light.
2. **Distance & Bandwidth Capacity**
    - **Ethernet (Copper):** Limited to a maximum effective distance of 100 meters (328 feet) per cable run without signal boosters, with lower total bandwidth capacity compared to light.
    - **Fiber Optic:** Transmits data across significantly longer distances—up to several kilometers or tens of kilometers without severe signal loss—at much higher bandwidth capacity.

3. **Interference Resistance**
    - **Ethernet (Copper):** Susceptible to electromagnetic interference (EMI) and radio frequency interference (RFI) from power lines, fluorescent lighting, and heavy machinery.
    - **Fiber Optic:** Completely immune to electrical interference because signal transmission relies entirely on light rather than electricity.

Real-World Use Cases:

- **Ethernet Cable (Copper): Home Wi-Fi Router Connection** — Connecting a gaming console, desktop PC, or smart TV directly to a home internet router using a standard Cat6 Ethernet cable for low-latency, short-distance networking.
- **Fiber Optic Cable: College Campus Backbone / Data Center Interconnect** — Linking separate buildings across a university campus or connecting servers across large data centers to handle massive volumes of network traffic over long distances without performance loss.

2.Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



3.Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Step-by-Step Guide: Attaching an RJ45 Connector to a CAT6 Cable

To attach an RJ45 connector to a CAT6 Ethernet cable using an RJ45 crimping tool and the T568B wiring standard.

Tools and Materials Required

- CAT6 Ethernet cable
- RJ45 connector
- RJ45 crimping tool
- Cable stripper/cutter
- LAN cable tester (recommended)

Procedure

Step 1: Cut the CAT6 Cable

Cut the CAT6 cable to the required length using the cutter on the crimping tool or a cable cutter. Make a clean, straight cut at the end.

Step 2: Remove the Outer Jacket

Use the stripping section of the crimping tool to carefully remove about 2–3 cm of the outer plastic jacket. Be careful not to damage the insulation around the individual wires.

Step 3: Separate the Wires

Inside the cable, there are four twisted pairs. Carefully untwist and separate the eight individual wires. Remove any plastic separator or rip cord if present.

Step 4: Arrange the Wires in T568B Order

Straighten the wires and arrange them from left to right in the following order:

Pin	Wire Colour
1	White-Orange
2	Orange
3	White-Green
4	Blue
5	White-Blue
6	Green
7	White-Brown
8	Brown

This color sequence should be checked carefully before inserting the wires into the connector.

Step 5: Straighten and Trim the Wires

Hold all eight wires tightly together in the correct order. Straighten them and cut the ends evenly so that all eight wires have approximately the same length.

Step 6: Insert the Wires into the RJ45 Connector

Hold the RJ45 connector with the contacts facing upward and the locking tab underneath. Carefully push all eight wires into the connector while maintaining the T568B order.

Push the wires far enough so that they reach the front of the connector and the outer cable jacket enters the connector far enough to provide strain support.

Step 7: Check the Wiring

Before crimping, check the connector again. Make sure:

- All eight wires are present.
- The color order is correct.
- The wires reach the front of the connector.
- The cable jacket is inside the connector.

Step 8: Crimp the Connector

Place the RJ45 connector into the appropriate socket of the crimping tool. Squeeze the handles firmly until the connector is properly crimped. The metal contacts should press down onto the individual wires and secure the connector to the cable.

Step 9: Inspect the Finished Cable

Remove the cable from the crimping tool and visually inspect the connector. Make sure the wires have not moved out of their correct positions and that the connector is firmly attached.

Step 10: Test the Cable

Connect the finished cable to a LAN cable tester. A correctly terminated straight-through cable should show the eight connections in the correct sequence from pin 1 through pin 8.

4.Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.

For a gaming café with 10 PCs, I would choose Ethernet (CAT6) cables and an RJ45 crimper. Ethernet is fast enough for LAN gaming, works well over the relatively short distances inside a café, and is much more affordable and easier to install than fiber optics. The RJ45 crimper can be used to attach connectors to the CAT6 cables, making it simple to create and repair network connections. Fiber optics and a fiber cleaver would be more suitable for very long distances or high-speed backbone connections.